

Does the following function $f(x_1, x_2) : \mathbb{R}^2 \rightarrow \mathbb{R}$ have a global maximum and a global minimum in the feasible set A ?

$$f(x_1, x_2) = x_1^2 + x_2^2 - 2x_1x_2$$

,

$$A = \{(x_1, x_2) \in \mathbb{R}^2 \mid (x_1 - 1)^2 + (x_2 - 2)^2 \leq 1\}.$$

Hint: use the Weierstrass extreme value theorem.